MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2026
Fourth Semester

DSC-A – MG4DSCZGY201
BIOLOGICAL CHEMISTRY

Duration: 1.5 hr Maximum Mark:50

PART A – MCQ (12 × 1 mark) Write all 12.
(12 X 1 =12 marks)

1. The complete and catalytically active form of an enzyme consisting of a protein part and a non-protein component is called: [K][CO:2]
   A. Apoenzyme
   B. Abzyme
   C. Holoenzyme
   D. Cofactor
2. Which type of cellular respiration occurs in the absence of oxygen? [K][CO:4]
   A. Aerobic respiration
   B. Anaerobic respiration
   C. Oxidative phosphorylation
   D. Krebs cycle
3. Amino acids exist as zwitterions at their: [K][CO:1]
   A. Highest pH value
   B. Lowest pH value
   C. Isoelectric point
   D. Denatured state
4. Atherosclerosis is characterized by: [K][CO:4]
   A. Accumulation of glucose in blood
   B. Thickening and hardening of arterial walls due to lipid deposits
   C. Breakdown of proteins in tissues
   D. Deficiency of cholesterol in arteries
5. Which of the following oligosaccharides is composed of glucose and galactose?
   A. Maltose [K][CO:1]
   B. Lactose
   C. Sucrose
   D. Cellobiose
6. Glycogenolysis is the process of: [K][CO:4]
   A. Formation of glycogen from glucose
   B. Conversion of glycogen into glucose
   C. Breakdown of glucose into pyruvate
   D. Conversion of glucose into fatty acids
7. Which of the following carbohydrates primarily serves as an energy storage molecule in plants? [K][CO:1]
   A. Cellulose
   B. Starch

C. Chitin
D. Sucrose

8. Which of the following best describes isoenzymes? [K][CO:2]
   A. Enzymes inhibited by feedback mechanisms
   B. Enzymes that require cofactors for activity
   C. Different molecular forms of the same enzyme catalyzing the same reaction
   D. Enzymes regulated by allosteric effectors
9. Rancidity of fats is mainly caused by: [K][CO:1]
   A. Hydrolysis of proteins
   B. Oxidation of fatty acids leading to free radical formation
   C. Polymerization of carbohydrates
   D. Reduction of triglycerides
10. Triglycerides are formed by the esterification of: [K][CO:1]
   A. One glycerol molecule with one fatty acid
   B. One glycerol molecule with three fatty acids
   C. Three glycerol molecules with one fatty acid
   D. Two glycerol molecules with two fatty acids
11. The primary function of the pentose phosphate pathway is to:
   A. Produce ATP directly [K][CO:4]
   B. Generate NADPH and ribose-5-phosphate
   C. Convert glucose into glycogen
   D. Oxidize fatty acids
12. Which of the following is considered a high-energy compound in cellular metabolism?
   A. Glucose-6-phosphate [K][CO:3]
   B. Adenosine triphosphate
   C. Pyruvate
   D. Lactate

PART B – Short Answer (3 marks) Write any 6.
(6 X3 =18 marks)

13. Draw the Fischer's and Haworth's structures of glucose [K][CO:1]

14. Explain the causes and symptoms of Diabetes Mellitus**.** [U][CO:4]

15. Describe how temperature and pH influence the activity of enzyme-catalyzed reactions. [U][CO:2]
16. Describe the difference between transamination and deamination in amino acid metabolism. [U][CO:4]
17. Detail lactic acid fermentation with its steps and significance. [U][CO:4]

18. Write a short note on coenzymes and their role in enzyme action**.** [U][CO:2]
19. Illustrate coupled reactions and explain their importance in bioenergetics. [U][CO:3]
20. Differentiate between nucleosides and nucleotides with examples. [U][CO:1]

PART C – Short Essay (5 marks) Write any 4.
(4 X5 =20 marks)

21. Compare and contrast reversible and irreversible enzyme inhibition in terms of mechanism and effect on enzyme function. [U][CO:2]

22. Discuss the Ramachandran plot and its significance in understanding protein conformation. [U][CO:1]

23. Explain the process of fatty acid activation and its significance in metabolism. [U][CO:4]
24. Distinguish enantiomers, anomers, and epimers and give one example of each. [U][CO:1]

25. Discuss the relationship between substrate concentration and reaction rate in enzyme kinetics. [U][CO:2]

26. Represent the Krebs cycle in a diagram, showing substrate-level phosphorylation and electron carrier formation**.** [U][CO:4]

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